

Cosets & Lagrange's Thm

Section 10

Examples of subgroups of finite groups we've seen:

- $\mathbb{Z}_n \cong \langle d \rangle \leq \mathbb{Z}_n$ for $d \mid n$ $\frac{n}{d}$ divides n
order n

- $A_n \leq S_n$ $|A_n| = \frac{n!}{2}$ divides $n! = |S_n|$
 $\frac{n!}{2}$ $n!$

- $D_n \leq S_n$ $|D_n| = 2n$ divides $n! = |S_n|$.

Cosets

Let $H \leq G$. We exhibit two partitions of G by defining equivalence relations on G :

↙ if and only if

$$\sim_L : a \sim_L b \text{ iff } a^{-1}b \in H$$

$$\sim_R : a \sim_R b \text{ iff } ba^{-1} \in H$$

Prop $\sim_L$ and $\sim_R$ are equivalence relations.

Pf: we prove for $\sim_L$. Exercise: check for $\sim_R$.

Reflexive: for any $a \in G$, to check $a \sim_L a$ we ask if $a^{-1}a \in H$. But $a^{-1}a = e \in H$ since H is a subgroup.

Symmetric: $\S a \sim_L b$. Then $a^{-1}b \in H$, so $(a^{-1}b)^{-1} \in H$. But $(a^{-1}b)^{-1} = b^{-1}a$. So $b \sim_L a$.

Transitive: $\S a \sim_L b$ and $b \sim_L c$, so $a^{-1}b, b^{-1}c \in H$.

Thus $(a^{-1}b)(b^{-1}c) = a^{-1}c \in H$. So $a \sim_L c$. $\square$

What do the cells of the corresponding partition look like?

$$a \sim_L x \iff a^{-1}x \in H, \text{ so } a^{-1}x = h \text{ for some } h \in H.$$

In other words, $x = ah$ for some $h \in H$.

Def: $aH = \{ah \mid h \in H\}$ ← a left coset

$$Ha = \{ha \mid h \in H\} \leftarrow \text{a right coset}$$

Note: aH is not a subgroup of G unless $aH = H$, which is true iff $a \in H$.



Ex: $H = 3\mathbb{Z} \leq \mathbb{Z} = G.$

"

$$\{\dots, -6, -3, 0, 3, 6, \dots\}$$

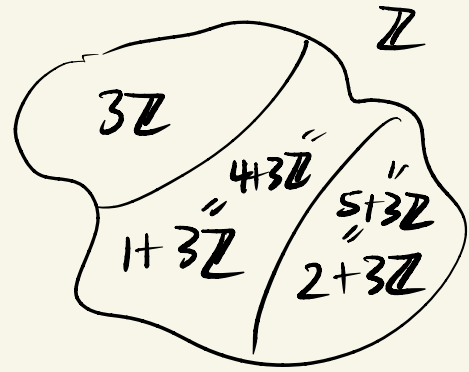
$$aH = a + H = \{a + h \mid h \in H\}$$

$$0 + H = H$$

$$1 + H = \{\dots, -5, -2, 1, 4, 7, \dots\}$$

$$2 + H = \{\dots, -4, -1, 2, 5, 8, \dots\}$$

$$3 + H = \{\dots, -3, 0, 3, 6, 9, \dots\} = H$$



$$0 \sim_L 3 \text{ since } -0 + 3 \in H$$

$$1 \sim_L 4 \text{ since } -1 + 4 \in H$$

Ex: $A_n \leq S_n$.

$$eA_n = A_n$$

$$(12)A_n = \{\text{odd permutations of } n\} = S_n \setminus A_n$$

"
 σA_n for any odd permutation σ

Ex: $\langle (12) \rangle \leq S_3$

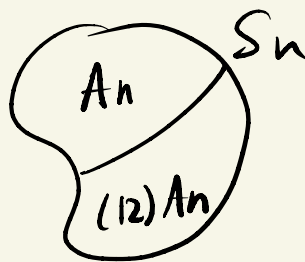
"
 H

left cosets

$$H = \{e, (12)\}$$

$$(23)H = \{(23), (23)(12)\} \neq$$

$$(13)H = \{(13), (13)(12), (123)\}$$



right cosets

$$He = H = \{e, (12)\}$$

$$H(23) = \{(23), (12)(23)\}$$

"
 (123)

$$H(13) = \{(13), (132)\}$$

If G is abelian then $\forall a \in G$

$$aH = \{ ah \mid h \in H \} = \{ ha \mid h \in H \} = Ha$$

Prop: Every (left or right) coset of $H \leq G$ has the same number of elements as H .

Pf: let $a \in G$. Want to give a bijection from H to aH .

Define $f: H \rightarrow aH$ by $h \mapsto ah$.

Injective: For any $b \in aH$ we know $ah = b$ has a unique solution in G .

Surjective: let $b \in aH$. By definition, $b = ah$ for some $h \in H$. So $f(h) = b$.

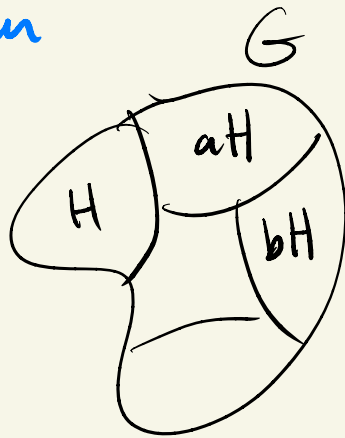


Thm (Lagrange's Theorem) let G be a finite group and $H \leq G$, then $|H|$ divides $|G|$.

Pf: Since the cosets partition G , and all cosets have size $|H|$, if there are r distinct cosets, then

$$|G| = r \cdot |H|.$$

□



Cor: Every group of prime order is cyclic.

Pf: Let G be a group of order p , with p prime. Let $a \neq e$ be any non-identity element of G . We claim $\langle a \rangle = G$. We know $\langle a \rangle$ is a subgroup of G . So $|\langle a \rangle|$ divides $|G| = p$ by Lagrange's Thm. But $|\langle a \rangle| \geq 2$ since $e, a \in \langle a \rangle$, so it must be $|\langle a \rangle| = p$, so $\langle a \rangle = G$. $\square$

Cor: Every group of prime order is abelian.

Cor: The order of an element of a finite group divides the order of the group.

Pf: Lagrange's Thm implies $|\langle a \rangle|$ divides $|G|$.

The order of $a \in G$ is the order of $\langle a \rangle$, in other words, its the smallest $r > 0$ with $a^r = e$.

Cor: If G is any finite group, of order n . And $a \in G$.

Then $a^n = e$.

Pf: By the corollary, $a^d = e$ for some $d \mid n$, namely $d = \text{order of } a$.

So $d \cdot r = n$ for some $r \in \mathbb{Z}_{>0}$, so $a^n = (a^d)^r = e^r = e$. $\square$

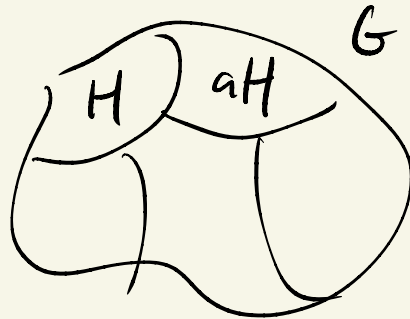
Def: Let $H \leq G$. The number of ^{distinct} left cosets of H in G is called the **index** $(G:H)$ of H .

Ex: $H = 3\mathbb{Z} \leq \mathbb{Z}$

$$(\mathbb{Z} : 3\mathbb{Z}) = 3$$

Ex: G finite. Then

$$(G:H) = \frac{|G|}{|H|}$$



$$\mathbb{Z}_n^{\times} = \{ i \mid 1 \leq i \leq n-1 \text{ and } \gcd(i, n) = 1 \}$$

Prop: $\mathbb{Z}_n^{\times}$ is a group under multiplication mod n .

Pf:

Identity: $1 \in \mathbb{Z}_n^{\times}$ since $\gcd(1, n) = 1$.

Closure: if i and j are relatively prime to n , then so is ij .

Assoc: ✓

Inverse: If $\gcd(i, n) = 1$, then we can write
 $(a, b \in \mathbb{Z}) \quad ai + bn = 1 \xrightarrow{\text{mod } n} ai \equiv 1 \pmod{n}$, so a gives inverse for i . $\square$

If $n=p$ is prime:

$\mathbb{Z}_p^\times = \{1, 2, \dots, p-1\}$ is a group of order $p-1$.

Thm (Fermat's Little Theorem).

Let $a \in \{1, 2, \dots, p-1\}$. Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Pf: $a^{|\mathbb{Z}_p^\times|} = e \in \mathbb{Z}_p^\times$

□

so $a^{p-1} \equiv 1 \pmod{p}$.